# How Much Will Be Charged?

## A Sparse Time-Series Forecasting Approach for EV Charging Energy

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KEBA AI Coding Contest 2026

June 1, 2026

# Problem Statement

**Goal:** predict the hourly energy charged for each user/charger in the test period.

Each row contains:

- `unique_id`: anonymized charger/user identifier
- `ds`: hourly timestamp
- `y_e`: charged energy amount, the Level 2 target
- `y_b`: binary charging indicator available in training

**Evaluation metric:** Mean Squared Error (MSE).

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Because MSE squares errors, missing a high-energy charging session is very expensive.

# Data Overview

| Item | Value |
|---|---|
| Training period | 2024-01-01 to 2025-03-31 |
| Test period | 2025-04-01 to 2025-05-31 |
| Number of users/chargers | 15 |
| Training rows | 164,160 |
| Test rows | 21,960 |
| Time granularity | Hourly |
| Target | Energy charged per hour |

The data is a panel time series: many users observed at the same hourly timestamps.

# Main Observation: The Target Is Sparse

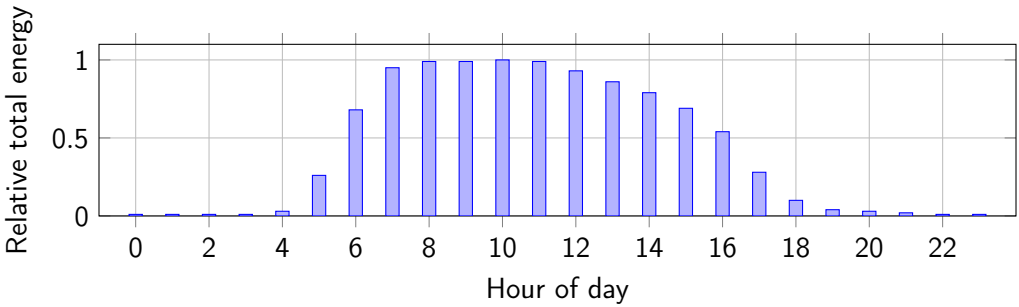
Most hourly rows have zero charged energy.

| Quantity | Approximate value |
|---|---|
| Share of non-zero energy rows | 5.76% |
| Share of zero-energy rows | 94.24% |
| Mean hourly energy | 150.13 |
| Maximum hourly energy | 8803.44 |

**Insight:** this is not a normal dense regression problem. It is mostly a "when will charging happen?" problem, followed by "how much energy if it happens?"

# Energy Is Concentrated in Business Hours

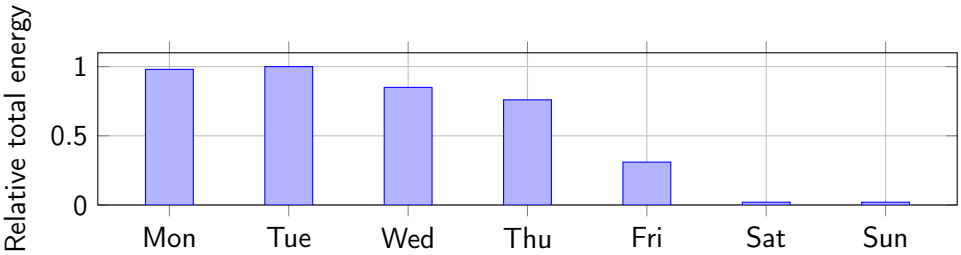
The strongest time-of-day signal is that charging mainly occurs during working hours.




**Feature implication:** hour-of-day and user-hour behavior are essential.

# Weekday Pattern

Charging is much more common from Monday to Thursday, with Friday lower and weekends almost inactive.




**Feature implication:** weekday and user-weekday-hour interactions are important.

# Users Behave Very Differently

The anonymized users have very different charging intensities.

- Some users charge frequently with many active days.
- Some users stop charging during the training period.
- One user appears only near the end of March.
- User-level behavior is more predictive than global averages.

**Feature implication:**

- `unique_id`
- user-hour averages
- user-weekday-hour averages
- recent activity / days since last charge

# Modeling Strategy

I treated the prediction as expected hourly energy:

$$\mathbb{E}[\text{energy}] = P(\text{charging happens}) \times \mathbb{E}[\text{energy} \mid \text{charging happens}]$$

This is useful because:

- Most hours are zero.
- The binary charging event is the hardest part.
- Positive charging amounts have user-specific patterns.

The final solution is an ensemble of several interpretable components, followed by calibration and smoothing.

# Features Considered

| Feature group | Examples |
|---|---|
| Calendar features | hour, weekday, month, holiday/bridge day |
| User identity | `unique_id` |
| User-time behavior | user-hour mean, user-weekday-hour mean |
| Recent behavior | 30, 60, 90, 180, 365 day windows |
| Seasonal lags | 7, 14, 28, 364, 365, 371 day lags |
| Activity state | days since last non-zero charge |
| Positive-energy profile | average positive energy by user/hour |

# Component 1: Historical Aggregate Demand

The first component estimates expected energy directly from historical averages.

Examples:

- mean energy for the same user, weekday, and hour
- mean energy for the same user and hour
- recent 60/90 day behavior
- same weekday last year signal

Simplified form:

$$\hat{y}_{agg} = w_1 \cdot \text{user-dow-hour mean} + w_2 \cdot \text{recent user-hour mean} + w_3 \cdot \text{seasonal lag}$$

This gave a stable baseline but was too smooth by itself.

# Component 2: Probability Times Amount

The second component models the sparse structure explicitly.

$$\hat{y}_{pa} = \hat{P}(\text{charge at user, weekday, hour}) \times \widehat{\text{positive energy at user, hour}}$$

**Probability estimate:**

$$\hat{P} = \text{mean of } y_b \text{ for same user, weekday, hour}$$

**Amount estimate:**

$$\widehat{\text{amount}} = \text{mean positive } y_e \text{ for same user and hour}$$

This component improved the score because it respected the zero-heavy structure.

# Activity Gating

Some users stopped charging before the test period.

For each user, I calculated:

$$\text{days since last charge}$$

Then I down-weighted or removed users that had become stale.

**Why it helped:**

- Historical averages alone keep predicting old users.
- If a charger has been inactive recently, April-May demand is likely lower.
- This reduced false positives on inactive users.

# Calendar and Holiday Corrections

The data looked like workplace charging, so public holidays and bridge days mattered.

I reduced predictions on:

- Easter Monday
- Labour Day
- Ascension Day
- likely bridge days after holidays

This correction was kept mild in the final model because overly aggressive holiday suppression hurt some submissions.

# Final Ensemble

The final model combined multiple components:

1. historical aggregate demand
2. activity-gated holiday component
3. probability-times-amount component
4. public-line calibrated component
5. seasonal lag probe
6. new-user probe

The final clean component was:

$$\hat{y}_{clean} = \hat{y}_{span} + \sum_{k} c_k \left( \hat{y}_k - \hat{y}_{span} \right)$$

where each component added a different view of the same demand pattern.

# Why Smoothing Was Needed

One experiment sharpened the predictions by pushing more energy into the highest predicted hours.

That made the score worse.

**Interpretation:**

- The model was already concentrated enough.
- Over-confident high predictions increased squared error.
- A smoother distribution was better calibrated.

Final smoothing:

$$\hat{y}_{final} = \hat{y}_{clean} + \alpha\,(\hat{y}_{sharp} - \hat{y}_{clean})$$

with negative $\alpha$, meaning the model moves away from the sharpened prediction.

# Result

Final submitted model:

$$\boxed{\text{MSE} = 385028.9067}$$

The final output file had the required format:

```
unique_id; ds; y_hat
```

# What Worked

- Recognizing the target as sparse, not ordinary dense regression.
- Separating charging probability from positive energy amount.
- Using user-specific hour and weekday patterns.
- Suppressing recently inactive users.
- Keeping holiday correction mild instead of extreme.
- Calibrating the final prediction distribution with smoothing.

# What Did Not Work

- Pure same-period-last-year replay was too brittle.
- Predicting exact discrete charging days overfit local patterns.
- Aggressively adding a new-user weekly routine hurt the score.
- Over-sharpening high-confidence hours increased MSE.
- A black-box LightGBM attempt overpredicted almost every row.

These failures were useful because they showed that the final model needed calibrated expected energy, not over-confident event reconstruction.

# Final Takeaway

The core idea was:

**Forecast expected charging energy by combining**
**when charging is likely**
**with how much energy is typical.**

The final score came from an interpretable ensemble of user-time patterns, activity state, calendar effects, and smoothing calibration.

# Thank You

Questions?